

CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

OBJECTIVES:

- i) To generate an Amplitude Modulated (AM) signal in MATLAB using specified carrier and modulating signal parameters.
- ii) To determine and verify the upper sideband (USB) and lower sideband (LSB) frequencies of the AM signal using frequency spectrum analysis in MATLAB.
- iii) To calculate the carrier power, sideband power, and total transmitted power of the AM signal for a given modulation index using MATLAB.
- iv) To analyze the effect of varying the modulation index on sideband power distribution and overall transmission efficiency through MATLAB simulation.

PROGRAM:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Parameters
```

```
Am = 1;           % Message amplitude
```

```
fm = 500;         % Message frequency (Hz)
```

```
Ac = 5;           % Carrier amplitude
```

```
fc = 5000;        % Carrier frequency (Hz)
```

```
fs = 20 * fc;     % Sampling frequency
```

```
%% Time Vector
```

```
t = 0:1/fs:4/fm;
```

```
%% Objective 1: Generate AM Signal
```

```
m = Am * sin(2*pi*fm*t);
```

```
c = Ac * sin(2*pi*fc*t);
```

```
mu = Am / Ac;           % Modulation Index
```

```
am = Ac * (1 + mu*m) .* sin(2*pi*fc*t);
```

```
% Demodulation
```

```
dem = abs(hilbert(am));
```

```
dem = dem - mean(dem);
```

```
%% Objective 2: Frequency Spectrum Analysis
```

```
N = length(am);
```

```
f = linspace(-fs/2, fs/2, N);
```

```
AM_FFT = abs(fftshift(fft(am)))/N;
```

```
USB = fc + fm;
```

```
LSB = fc - fm;
```

```
%% Objective 3: Power Calculations
```

```
Pc = Ac^2/2;           % Carrier Power
```

```
Psb = (mu^2 * Pc)/4;    % Sideband Power (Each)
```

```
Pt = Pc + 2*Psb;        % Total Power
```

```
Efficiency = (2*Psb/Pt)*100; % Transmission Efficiency
```

```
%% Display Results
```

```
fprintf('\n----- AM PARAMETERS ----- \n');
```

```
fprintf('Message Frequency    = %d Hz\n', fm);
```

```
fprintf('Carrier Frequency      = %d Hz\n', fc);
```

```
fprintf('Modulation Index (μ)    = %.2f\n', mu);
```

```
fprintf('\n----- Objective 2: Sideband Verification ----- \n');
```

```
fprintf('Upper Sideband (USB) = %d Hz\n', USB);
```

```
fprintf('Lower Sideband (LSB) = %d Hz\n', LSB);
```

```
fprintf('\n----- Objective 3: Power Calculation -----\n');
```

```
fprintf('Carrier Power      = %.2f W\n', Pc);
```

```
fprintf('Each Sideband Power = %.3f W\n', Psb);
```

```
fprintf('Total Power          = %.3f W\n', Pt);
```

```
fprintf('Transmission Efficiency = %.2f %%\n', Efficiency);
```

```
%% Objective 4: Effect of Modulation Index
```

```
mu_values = [0.2 0.4 0.6 0.8 1.0];
```

```
Power_SB = zeros(size(mu_values));
```

```
Power_Total = zeros(size(mu_values));
```

```
Eff = zeros(size(mu_values));
```

```
fprintf('\n----- Effect of Modulation Index -----\n');
```

```
fprintf('Mu\tSideband Power(W)\tTotal Power(W)\tEfficiency(%%)\n');
```

```
for i = 1:length(mu_values)
```

```
    mu1 = mu_values(i);
```

```
    Psb1 = (mu1^2 * Pc)/4;
```

```
    Pt1 = Pc + 2*Psb1;
```

```
    Eff1 = (2*Psb1/Pt1)*100;
```

```
    Power_SB(i) = Psb1;
```

```
    Power_Total(i) = Pt1;
```

```
    Eff(i) = Eff1;
```

```

fprintf('%0.1f\t%0.3f\t\t\t%0.3f\t\t%0.2f\n', ...
        mu1,Psb1,Pt1,Eff1);
end

%% Plots
figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1)
plot(t,m,'LineWidth',1.5)
title('Message Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

subplot(3,2,2)
plot(t,c,'LineWidth',1.5)
title('Carrier Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

subplot(3,2,3)
plot(t,am,'LineWidth',1.5)
title('AM Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

```

```

subplot(3,2,4)
plot(t,dem,'LineWidth',1.5)
title('Demodulated Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on

```

```

subplot(3,2,5)
plot(f,AM_FFT,'LineWidth',1.5)
xlim([fc-3*fm fc+3*fm])
title('Frequency Spectrum')
xlabel('Frequency (Hz)')
ylabel('Magnitude')
grid on

```

```

subplot(3,2,6)
bar([Pc Psb Psb Pt])
set(gca,'XTickLabel',{'Carrier','USB','LSB','Total'})
title('Power Distribution')
ylabel('Power (W)')
grid on

```

```

%% Effect of Modulation Index
figure('Name','Effect of Modulation Index','NumberTitle','off');

```

```

subplot(2,1,1)
plot(mu_values,Power_SB,'-o','LineWidth',2)
xlabel('Modulation Index (\mu)')
ylabel('Sideband Power (W)')

```

title('Sideband Power vs Modulation Index')

grid on

subplot(2,1,2)

plot(mu_values, Eff, '-s', 'LineWidth', 2)

xlabel('Modulation Index (μ)')

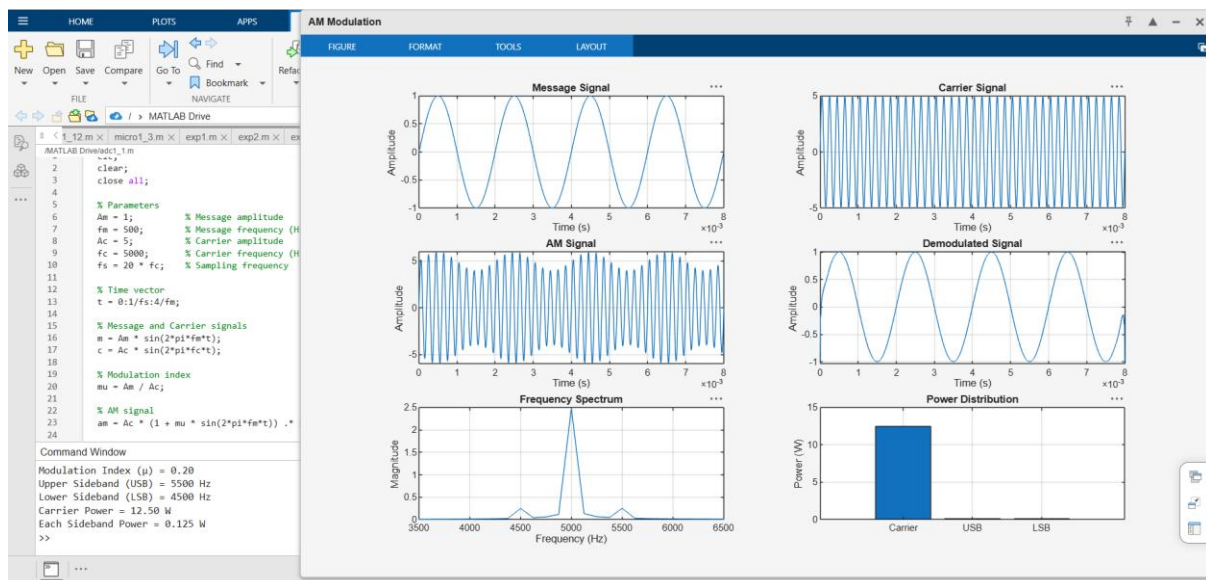
ylabel('Efficiency (%)')

title('Transmission Efficiency vs Modulation Index')

grid on

OBJECTIVE 1 - To generate an Amplitude Modulated (AM) signal in MATLAB using specified carrier and modulating signal parameters.

OUTPUT:



----- AM PARAMETERS -----

Message Frequency	= 500 Hz
Carrier Frequency	= 5000 Hz
Modulation Index (μ)	= 0.20

OBJECTIVE 2 - To determine and verify the upper sideband (USB) and lower sideband (LSB) frequencies of the AM signal using frequency spectrum analysis in MATLAB.

OUTPUT:

```
----- Objective 2: Sideband Verification -----  
Upper Sideband (USB) = 5500 Hz  
Lower Sideband (LSB) = 4500 Hz
```

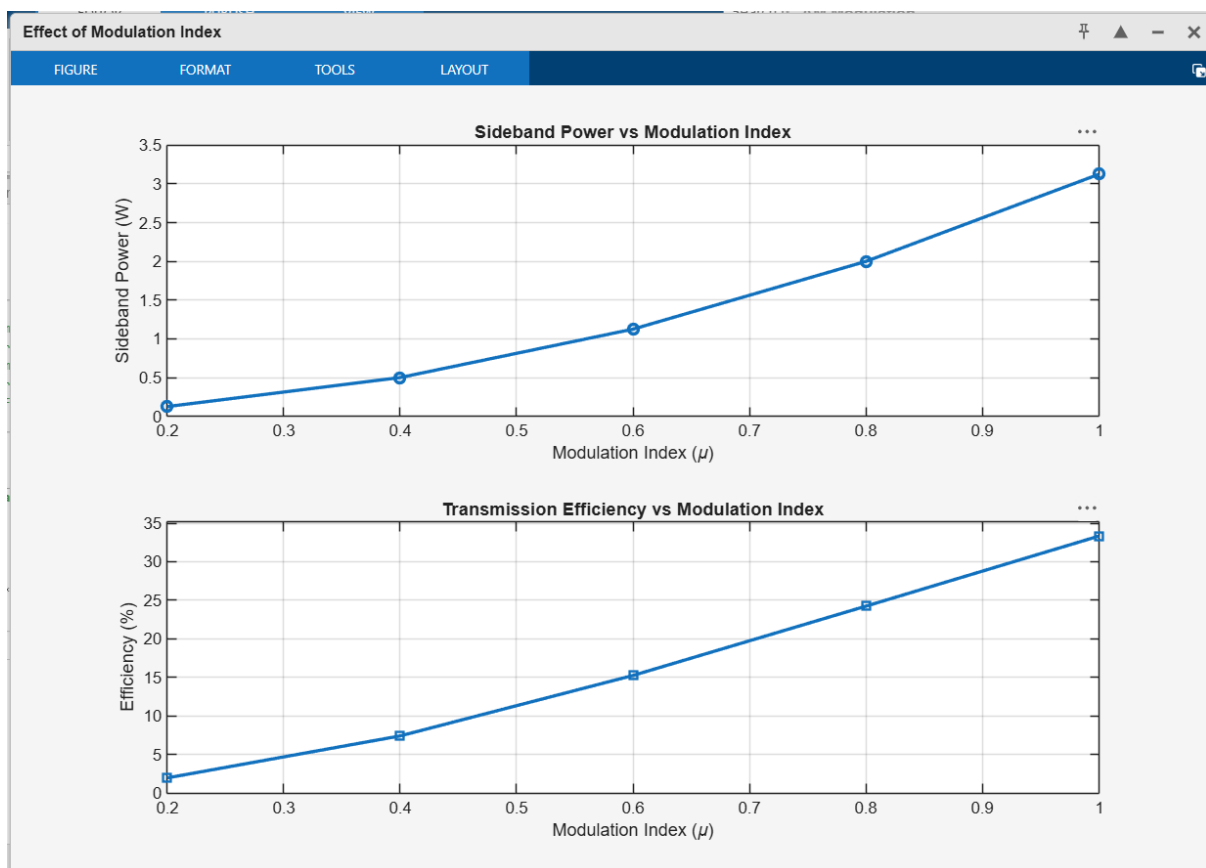
OBJECTIVE 3 - To calculate the carrier power, sideband power, and total transmitted power of the AM signal for a given modulation index using MATLAB.

OUTPUT:

```
----- Objective 3: Power Calculation -----  
Carrier Power      = 12.50 W  
Each Sideband Power = 0.125 W  
Total Power        = 12.750 W  
Transmission Efficiency = 1.96 %
```

OBJECTIVE 4 - To analyze the effect of varying the modulation index on sideband power distribution and overall transmission efficiency through MATLAB simulation.

OUTPUT:



```
----- Effect of Modulation Index -----
Mu      Sideband Power(W)      Total Power(W)  Efficiency(%)
0.2      0.125                  12.750          1.96
0.4      0.500                  13.500          7.41
0.6      1.125                  14.750          15.25
0.8      2.000                  16.500          24.24
1.0      3.125                  18.750          33.33
>> |
```